

System Design Document

Tower of Hanoi Software

Project Information

Project Name: Tower of Hanoi Software

Course: Software Development 1

Programming Language: Java

Development Methodology: Incremental Development

Document Version: 1.0

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1. Introduction

The Tower of Hanoi Software is a Java-based application designed to simulate the classic Tower of Hanoi puzzle. The software will provide an interactive graphical user interface where users can manually solve the puzzle while following the game rules. The application will also include an automatic solver based on the recursive Tower of Hanoi algorithm.

The purpose of this project is to demonstrate software development principles, object-oriented programming concepts, graphical user interface design, and algorithm implementation.

2. System Objectives

The objectives of the system are:

- Provide an interactive Tower of Hanoi game.
 - Allow users to move disks between towers.
 - Validate all user moves.
 - Prevent illegal moves.
 - Track player performance through move counting and timing.
 - Automatically solve the puzzle using recursion.
 - Display a winning message upon successful completion.
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3. System Architecture

The software follows a modular architecture where each component has a specific responsibility.

Main Components

Main Application Module

Responsible for launching the software.

User Interface Module

Responsible for displaying menus, buttons, towers, and disks.

Game Logic Module

Responsible for handling game rules and move operations.

Validation Module

Responsible for checking whether a move is legal.

Auto Solver Module

Responsible for solving the puzzle automatically using recursion.

Utility Module

Responsible for timer, move counter, restart functionality, and win detection.

4. Project Structure

towerofhanoi

- |— TowerOfHanoi.java
- |— MainMenu.java
- |— GameBoard.java
- |— Tower.java
- |— Disk.java
- |— GameController.java
- |— AutoSolver.java

This structure follows object-oriented programming principles where each class has a specific responsibility within the system.

5. Class Design

TowerOfHanoi.java

Starts the application.

MainMenu.java

Displays the main menu and navigation options.

GameBoard.java

Displays towers and disks during gameplay.

Tower.java

Represents a tower and stores disks.

Disk.java

Represents a disk object with a specific size.

GameController.java

Controls game flow, move validation, and game status.

AutoSolver.java

Implements the recursive algorithm used for automatic puzzle solving.

6. System Workflow

The system will operate according to the following workflow:

1. User launches the application.
2. Main Menu is displayed.
3. User selects "Start Game".
4. Game Board is loaded.
5. User moves disks between towers.
6. System validates each move.
7. Move counter and timer are updated.
8. If the puzzle is solved, the system displays a winning message.
9. User may restart the game or exit the application.

7. Functional Requirements

FR-01

The system shall allow the user to start a new game.

FR-02

The system shall display three towers.

FR-03

The system shall display disks of different sizes.

FR-04

The system shall allow disk movement between towers.

FR-05

The system shall validate all moves.

FR-06

The system shall prevent illegal moves.

FR-07

The system shall count the number of moves.

FR-08

The system shall display elapsed time.

FR-09

The system shall provide an automatic solver.

FR-10

The system shall detect puzzle completion.

8. Non-Functional Requirements

Performance

The software should respond to user actions immediately.

Usability

The interface should be simple and user-friendly.

Reliability

The software should correctly enforce all game rules.

Maintainability

The software should be modular and easy to update.

9. Future Enhancements

The following features may be added in future versions:

- Multiple difficulty levels
 - Sound effects
 - Scoreboard system
 - Dark mode interface
 - Save and Load game functionality
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10.Next Development Phase

In Week 3, graphical user interface (GUI) development will begin. The main application window, menu system, navigation controls, and user interaction components will be implemented using Java Swing.

11.Conclusion

The Tower of Hanoi Software is designed as a modular Java application that combines graphical user interface development, object-oriented programming, and recursive algorithm implementation. The system design provides a solid foundation for future development phases and ensures maintainability, scalability, and usability.